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Investigating Deep Learning Knowledge Transfer in Streamflow Prediction From Global to Local Catchment

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Key Points:

- LSTM models pretrained on data-rich regions significantly improves streamflow forecasting accuracy in data-scarce areas like Central Asia
- Grouping basins into Clusters based on catchment attributes helped to enhance model performance in target region
- The success of transfer learning depends on both the quantity and similarity of the training data to the target region

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Accurate streamflow prediction is critical for flood forecasting and water resource management, particularly in data-scarce regions. Deep learning models like Long Short-Term Memory (LSTM) offer a bridge to hydrologic regionalization utilizing climate data and catchment characteristics to improve behavioral insights and constrain predictive uncertainties. Here we evaluate transfer learning (TL) approaches using 441 "donor" basins from regions rich in quality hydrological data (Scotland GB-SCT; Switzerland CH; and Canada's British Columbia; BC) to pre-train LSTM runoff models by fine-tuning in data-poor areas like CA (36 target basins). Pairing measured streamflow (lagged) records with global climate data (ERA-5) boosted the explanatory power of LSTM predictions especially in snowmelt and glacier-influenced basins. A K-Means clustering algorithm was applied to categorize basins into five hydrologically meaningful Clusters (labeled 1–5) based on catchment attributes. The results show that TL-LSTM models perform better when pre-trained using Clusters compared to the locally trained model (NSE = 0.85 and KGE = 0.80). The LSTM model trained on basins Cluster 3 data ($LSTM_3$) most closely resembling those in the target region yielded the most accurate predictions. Fine-tuning with limited local data substantially improved prediction accuracy in the validation split (blind-tested and treated as "ungauged"), evidencing that even short-records of local data can enhance a regional model of hydrological behavior. These results demonstrate that TL-LSTM can effectively enhance streamflow prediction in data-scarce regions. These insights advance understanding of cross-basin generalization and support the development of efficient, scalable modeling strategies for hydrological prediction in regions with limited observational data.

Plain Language Summary LSTMs can improve streamflow predictions where data is limited due to their ability to transfer knowledge gained from data rich regions. We evaluated this transfer learning ability of LSTMs by first training on data-rich regions (Scotland, Switzerland, and British Columbia) and then fine-tuning for data-poor region like CA. LSTMs showed improved predictions, especially in basins characterized by snow and glaciers. We grouped basins into five clusters based on their catchment characteristics. We found that models trained on specific clusters and then finetuned in target region performed better than those trained only on local data. The best results came from the model trained with basins in Cluster 3, while including all basins during training did not always improve predictions over target region. Furthermore, we trained and fine-tuned TL-LSTMs using short periods of data from half of the basins in the target region. The other half were treated as truly ungauged to evaluate model performance. The results show that even a small amount of local data can significantly improve predictions in nearby ungauged basins. These results show that transfer learning can improve streamflow predictions in data scarce regions by training model from similar regions to help in flood management and water resources.

1. Introduction

Accurate prediction of streamflow across diverse watersheds is essential for effective flood forecasting and water resources management (Sattari et al., 2025). Advanced machine learning (ML) techniques are increasingly being used to learn complex, time-dependent relationships directly from data by identifying patterns embedded in meteorological forcing and physiographic characteristics (Foroumandi & Moradkhani, 2025; Nearing et al., 2021). Among these methods, Long Short-Term Memory (LSTM) networks have shown strong potential for capturing nonlinear hydrological dynamics (Kratzert et al., 2018, 2019). However, the performance of LSTMs can be severely compromised in basins with short, sparse, or incomplete observational records (Feng et al., 2020; Sattari & Moradkhani, 2025). To overcome this limitation, transfer learning (TL) can be employed by pre-training LSTM models on data-rich basins, which can then be fine-tuned on smaller, target data sets to adapt to local

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catchment characteristics (Gauch et al., 2021; Ma et al., 2021). This approach enables robust streamflow prediction in data-scarce or ungauged basins while reducing overfitting and improving generalization (Fang et al., 2022; Liu et al., 2023). TL offers a pathway to extend the benefits of ML based hydrological prediction to regions where direct training is not feasible (Khoshkalam et al., 2025; Yao et al., 2023).

Global hydrological stations are unevenly distributed, with sparse hydrometeorological networks in developing countries and denser ones in developed nations (Ma et al., 2021). Many regions outside the US and Europe encounter difficulties in formulating robust risk-based models of water resource responses due to limited streamflow gauge networks and a lack of adequate climate, physiographic and land use/land cover data. Additionally, in some regions such as Central Asia (CA), hydrological observations are not publicly accessible due to policy restrictions (Feng et al., 2021), which poses a considerable challenge in accurate streamflow predictions (Tsai et al., 2021). The Global Runoff Data Center (GRDC) provides evidence (http://grdc.bafg.de) that many basins worldwide have less than 3 years of daily streamflow data (Alipour & Kibler, 2018). This absence or scarcity of station data increases the risk of water-related disputes due to unreliable flow data especially in large-scale transboundary basins (Sharma et al., 2021; Sood & Smakhtin, 2015).

Predicting basin-scale streamflow is challenging due to complex environmental dynamics and inherent uncertainties, with model performance largely limited by the availability of high-quality data (Ghaneei & Moradkhani, 2025). Physically or process-based models use equations to represent the pathways and dynamics of water flow through the landscape. A key issue constraining their performance is the availability of observed data to calibrate and validate their representation of the real world (Khoshkalam et al., 2025; Ougahi & Rowan, 2025). In the absence of such data, streamflow predictions may be unreliable especially in data-scarce regions (Clark et al., 2015, 2017). Process-based models calibrated on a single catchment or a small group of catchments, often face challenges of transferability even within similar climate zones (Arsenault & Brissette, 2014; Guo et al., 2021). In this context, alternative modeling approaches, such as data-driven models or machine learning-based models (ML), may be used to complement or improve the predictions in the absence of extensive data (Feng et al., 2021; Kratzert et al., 2018; Liang et al., 2018). ML-based approaches have the advantage of requiring less domain-specific knowledge as evidenced in recent studies for flood prediction, emphasizing their ability to handle complex and nonlinear relationships in large data sets (e.g., Gauch et al., 2021; Kratzert et al., 2018; Nearing et al., 2024; Nevo et al., 2022). These models are using regional and global remote sensing and reanalysis products and emerging as practical alternatives to traditional hydrological modeling approaches (Kratzert et al., 2019; Ma et al., 2021; Rasheed et al., 2024).

ML models vary from simple regression models to large neural networks with numerous parameters. ML methods such as Artificial Neural Networks (ANNs) and Recurrent Neural Networks (RNNs) have been widely studied for hydrological modelling (Zhu et al., 1994). A deep learning model, the Long Short-Term Memory (LSTM) neural network (Hochreiter & Schmidhuber, 1997) has gained significant attention for time-series forecasting due to its ability to capture long-term dependencies in hydrological processes (Kratzert et al., 2018; Liang et al., 2023). Numerous studies have demonstrated the effectiveness of LSTM in predicting dynamic hydrological variables such as floods, runoff, and soil water content (Xu et al., 2023; Yao et al., 2023), but its application requires a large amount of training data (Bai et al., 2021). Furthermore, LSTM models trained on large and diverse data sets exhibit greater transferability, making them particularly effective in data-scarce regions (Zhang et al., 2024). However, the quality and quantity of data are crucial for training deep learning models, and the lack of flow measurements can lead to unreliable representations of the rainfall–runoff relationship (Peng et al., 2022; Rahmani et al., 2021).

LSTM models have been applied to streamflow modelling and regionalization, showing substantial advancements compared to previous state-of-the-art methods (Kratzert et al., 2019), largely due to their ability to capture temporal dependencies including lag effect in runoff. Reed et al. (1975) demonstrated that such a lag is a common characteristic of rainfall–runoff behavior across numerous catchments. This lag results from variations in surface characteristics that create a time delay between precipitation and runoff events (Nippgen et al., 2016). Accounting for such lag effects is particularly important for hydrological models like LSTMs, whose performance can be further enhanced through data integration (DI) of recent observations such as discharge or related variables, to improve near-term forecast accuracy (Feng et al., 2023; Li et al., 2024; Nearing et al., 2024). Previous studies have demonstrated that including lagged discharge observations as inputs to LSTM models can significantly enhance basin-scale forecasting (Khoshkalam et al., 2023; Ma et al., 2024; Mangukiya et al., 2023). For example,

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Feng et al. (2021) applied the LSTM model with various data integration techniques for streamflow prediction across the Continental United States. They demonstrated that DI not only improved model performance but also provided valuable hydrological insights.

The application of LSTM-based transfer learning (TL-LSTM) for streamflow prediction is still in its early stages of development (Bao et al., 2024; Rasheed et al., 2024). TL offers significant potential for enhancing predictions in data-scarce basins (Gauch et al., 2021; Ghaneei et al., 2024). Previous studies often focused on a single region or hydrologically similar catchments (Chen et al., 2024; Molina et al., 2025), yet only a few have investigated transferring knowledge across distinct regions or continents (Chen et al., 2024; Ma et al., 2024; Zhang et al., 2024). This study advances TL by proposing and evaluating a TL-LSTM framework that combines large-sample pretraining with fine-tuning at target basins. The availability of CAMELS-type data sets worldwide provides an opportunity to answer insufficiently explored questions such as (a) to what extent can knowledge learned from well-observed regions be reliably transferred to hydrologically distinct, data-sparse basins? (b) does increasing the number of donor basins during training consistently improve hydrological model performance, or offer limited or even degrades prediction? (c) does fine-tuning under data scarce scenario can improve prediction in nearby ungauged basins? Addressing these questions provides critical insights into the design of efficient and robust TL-LSTM strategies for streamflow prediction in data-limited regions. The significance of this work lies in exploring the potential of TL-LSTM to enhance streamflow predictions in one of the world's most vulnerable and water-stressed region such as CA.

2. Materials and Methods

2.1. Study Area

Comparative hydrological data were compiled from basins in four montane regions across the northern hemisphere, inclusive of Switzerland (CH; 247 basins), Scotland (GB-SCT; 160 basins), and British Columbia in Canada (BC; 33 basins); along with 36 basins from Central Asia (CA), see Figure 1. Basins from data-rich regions, such as CH, GB-SCT, and BC, serve as "Donor" catchments used for training models. In contrast, basins from data-scarce region, such as CA, are designated as "target" catchments, where model performance is evaluated under limited data conditions. The size of basins ranged from 4 km^2 to 25,791 km^2, with a median area of 330 km^2. Each region features variable temporal length and number of basins, depending on the availability of reliable hydrological long-term streamflow records. The three training regions feature a diverse range of hydroclimatic settings and a wide range of catchment characteristics spanning mountainous area to lowland plains. Taken together they provide a significant sample of mid-latitude upland environments ranging from humid maritime temperate to semi-arid cold continental.

The radar plots illustrate distinct climatic characteristics across catchments (e.g., GB-SCT, CH, BC, and CA) based on average monthly temperature (temp), potential evapotranspiration (pet), and precipitation (pre) (e, f and g respectively; Figure 1). CH catchments show the highest variability across all three variables, reflecting the complex topography and broad range of climatic conditions from lowland valleys to Alpine zones. BC catchments also display moderate to high variability, especially in pet and pre, influenced by both coastal and interior climates. GB-SCT catchments, in contrast, exhibit relatively moderate and uniform climatic conditions, with limited variability in temp and pet, consistent with its maritime climate. Climate variability remains the lowest and most uniform in CA, highlighting its cold, arid to semi-arid conditions and continental climate influence. Lower pet values in CA than in CH and BC show some variability, possibly due to elevation differences and aridity. Overall, the strong climatic contrasts exist among these regions, with CH being the most diverse and CA the most uniform. These differences have important implications for LSTM-based TL in streamflow prediction, as models trained on data-rich, variable regions (inclusive of systems which are poor analogues for the target systems) may struggle when applied to data-scarce regions (Chen et al., 2024; Xu et al., 2023).

2.2. Data Sets

Google Earth Engine (GEE) is a free cloud-based platform offering access to extensive geospatial data sets. A key computational challenge lies in processing gridded meteorological and attribute data, which was addressed using GEE via Python Application Programming Interface (API). We used meteorological forcing data (e.g., pre, temp and pet) from ERA5-Land, a data set for the climate reanalysis (fifth generation) of the European Center for Medium-Range Weather Forecasts (ECMWF), providing atmospheric variables with global coverage at ~9 km

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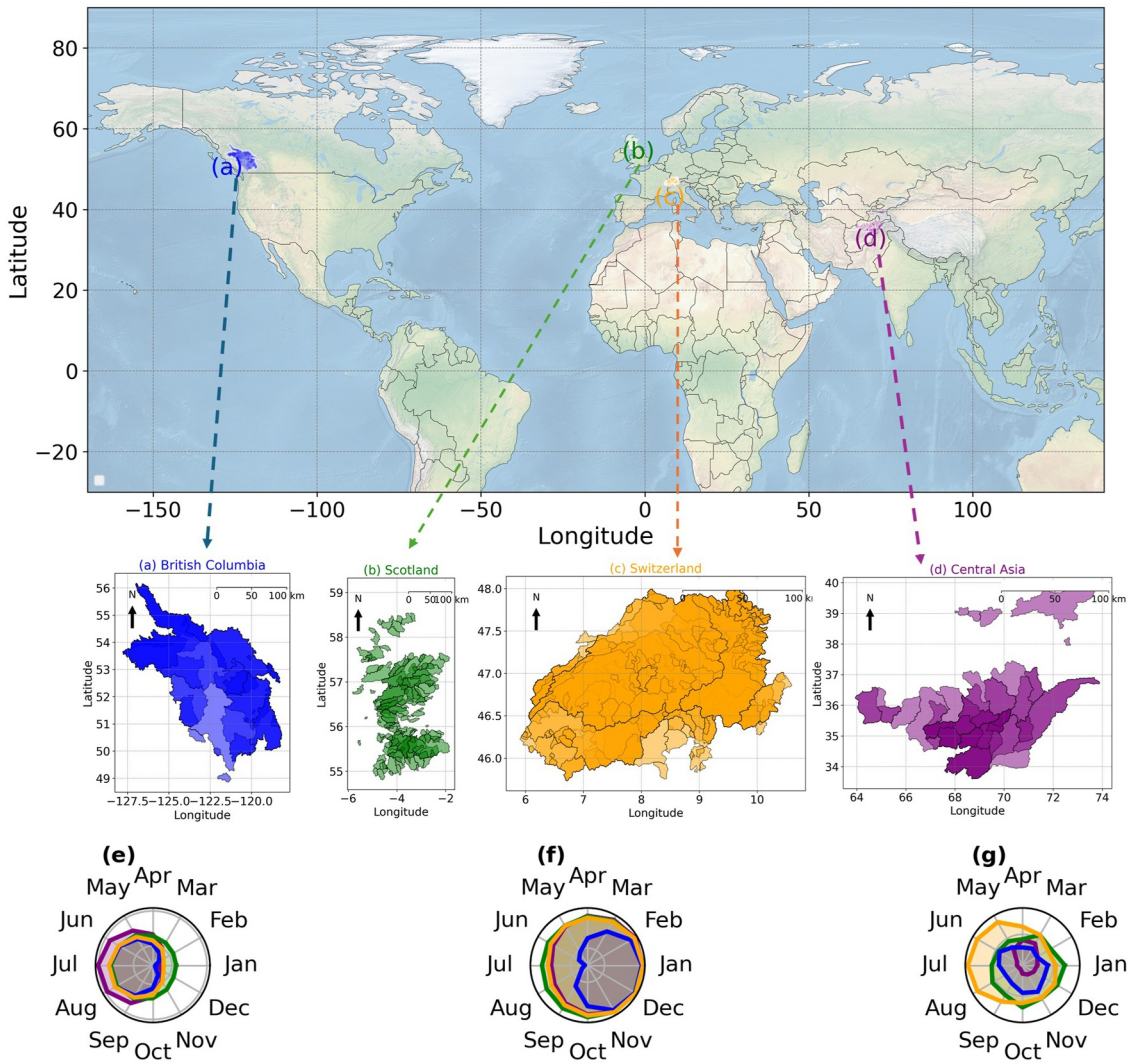


Figure 1. The geographical foundation of the study is depicted with small letters ((a): blue, (b): green, (c): orange, and (d): purple) representing the locations of regional catchments on the global map, along with detailed basin networks of donor regions: (a) British Columbia, Canada [BC], (b) Scotland, Great Britain [GB-SCT], (c) Switzerland [CH], and target catchments in panel (d) Central Asia [CA]. Three Taylor plots present the monthly averages of (e) temperature, (f) potential evapotranspiration, and (g) precipitation, with color-coded circles denoting each region: BC in blue, GB-SCT in green, CH in orange, and CA in purple.

resolution (Muñoz-Sabater et al., 2021) available in GEE at hourly time steps. For meteorological variables of each catchment, we derived area-weighted spatial averages for forcing variables and computed daily statistics from hourly data. LSTM-based hydrological modeling depends not only on meteorological inputs but also on catchment characteristics providing local context informing behavioral responses based on unique static properties of each catchment (*cf.* Arsenault et al., 2023; Kratzert et al., 2019). By relying on ERA5-Land, we ensured consistent meteorological inputs across all basins, minimizing discrepancies that can arise from combining regional data sets, many of which contain gaps or incomplete records over the study period.

We extracted basin descriptors from the study regions using HydroATLAS because of its compatibility with other HydroSHEDS products providing a standardized set of hydro-environmental attributes at a high spatial resolution worldwide (Linke et al., 2019). The data set includes 56 hydro-environmental variables, broken down into 281 individual attributes and grouped into six categories such as hydrology, physiography, climate, land cover, soil and geology, and anthropogenic factors. Climate attributes like evapotranspiration and climate moisture index play a crucial role in distinguishing regions into wet and dry. Similarly, hydrology, land use and terrain characteristics also help to differentiate basins based on discharge pattern, land cover and topography.

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Discharge data were obtained from multiple donors, including the Global Runoff Data Center (GRDC), the Global Streamflow Indices and Metadata Archive (GSIM), and CAMELS data sets (Catchment Attributes and MEteorology for Large-sample Studies) for specific countries, namely CAMELS-CH (Switzerland) and CAMELS-GB (Great Britain). CAMELS-CH is a comprehensive hydro-meteorological data set for hydrology containing 331 basins within Switzerland and neighboring countries like France and Italy (Höge et al., 2023). The diverse landscape of Switzerland ranges from mountainous regions and karst formations to heavily cultivated areas, offer a wide variety of hydrological regimes, including glacier-fed, snow-dominated, and rain-dominated catchments. CAMELS-GB is a comparable hydrology data set for Great Britain providing daily time series of river flow and meteorological data alongside catchment attributes from the UK National River Flow Archive (Coxon et al., 2020). It spans 671 catchments including GB-SCT, encompassing a diverse range of climatic, hydrological, landscape, and human management characteristics. The discharge time series from CAMELS-GB and CAMELS-CH were used only for basins with continuous records spanning 1980 to 2015.

Global streamflow data collections, such as the GSIM and GRDC, offer monthly and seasonal streamflow indices for over 35,000 locations. The GRDC (https://www.bafg.de/GRDC) also provides river discharge estimates, drawing from data collected at over 10,000 stations. However, the quality and duration of records vary significantly across individual basins (Kratzert et al., 2023). Streamflow data for BC and CA basins were downloaded from the GRDC portal.

2.3. Methodology

The flow diagram illustrates a two-step methodology for streamflow prediction using data integration and TL, aimed at improving performance in data-scarce region (Figure 2). In the first step, best-performing input variables such as lagged flow, temperature (temp), precipitation (pre) and potential evapotranspiration (pet) combination for LSTM model were identified. In the second step, based on best combination of input variables, TL-LSTM models were developed and compared with regionally trained LSTM model.

In the first step, preliminary experiments were conducted to investigate the performance of LSTM with different sets of forcing variables in both donor (CH, GB-SCT and BC) and target basins (CA) before clustering, inclusive of "lag" streamflow, pre, temp and pet (Table 1). Three LSTM model variants based on data from each group (Group-I, Group-II and Group-III) were developed for each region. In Group-I, lag observations of streamflow were utilized to predict streamflow to aid the model in capturing past flow conditions. Lagged streamflow values are taken at 1, 2- and 3-day (i.e., previous day), demonstrating how past observations contribute to future forecasts. In Group-II, LSTM model was trained with meteorological variables only (e.g., temp, pre, and pet). This analysis sought to reveal the influence of input variables as predictors for streamflow simulation. In Group-III, the LSTM model was trained using a combination of meteorological variables and lagged streamflow data. This approach allows algorithms to incorporate both meteorological inputs (pre, temp and pet) and streamflow observations (lagged streamflow) for improved streamflow predictions.

In the second step, a TL approach was applied, where six TL-LSTM models were developed (e.g., five trained separately on data from each cluster (Clusters 1 to 5) and one additional model trained on the combined data from all clusters). For example, the LSTM model pretrained using donor basins in Cluster 1 and fine-tuned on the target basins (Cluster 4) was named as $LSTM_1$. Similarly, the model pretrained on donor basins in Cluster 2 and fine-tuned on the target basins was named as $LSTM_2$, while the model pretrained using basins in Cluster 3 and fine-tuned on the target basins was named as $LSTM_3$, and LSTM model pretrained on Cluster 5 and fine-tuned on the target basins was named as $LSTM_5$. The base model, trained using only the data from basins within Cluster 4, is referred to as $LSTM_4$. This model serves as a regional benchmark, capturing hydrological patterns specific to Cluster 4 without any transfer learning from other regions. Lastly, the LSTM model pretrained on all basins (441 basins) and fine-tuned on the target basins was named as $LSTM_6$. Furthermore, to examine the impact of data volume on TL, LSTM models were (e.g., Cluster 1 to Cluster 3 and Cluster 5) pretrained on either a subset of basins or the complete data set from each cluster and subsequently transferred to target basins (Cluster 4) for prediction.

2.4. Principal Component Analysis and Clustering Evaluation

In regional hydrology studies, similarities and differences between basins are commonly assessed based on factors such as geographic proximity, climate, catchment size, physiography, hydrogeology, and land use/land

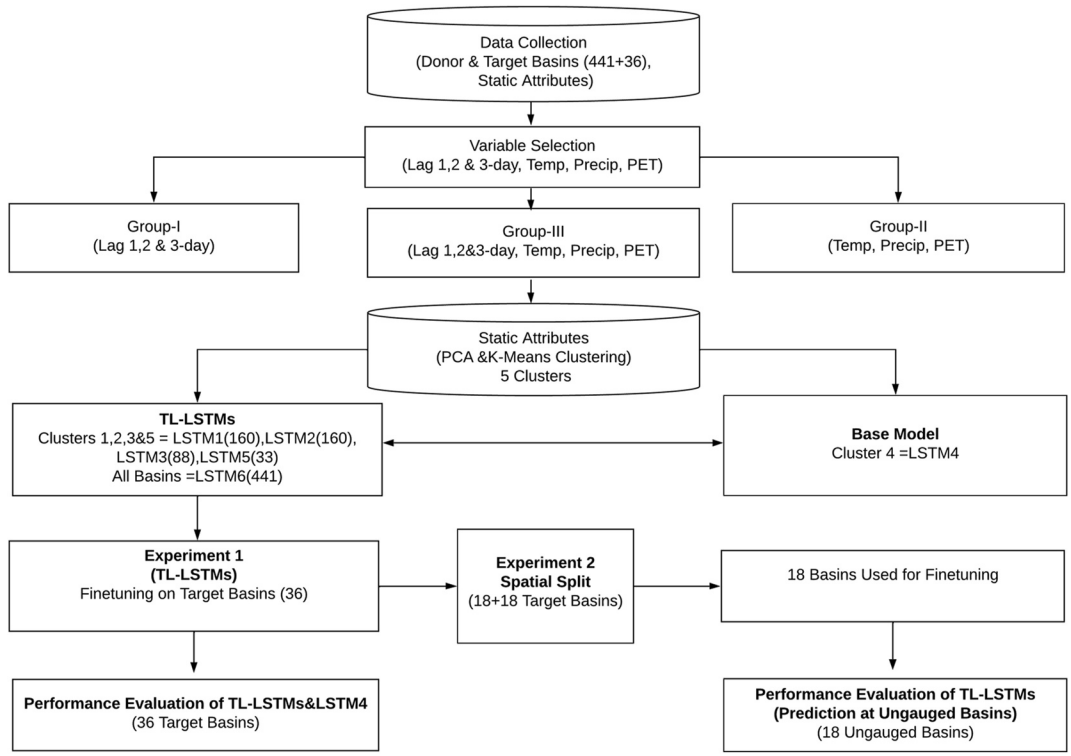


Figure 2. Methodological framework of transfer-learning for LSTM-based (TL-LSTM) regional hydrological modeling and prediction in ungauged basins. This framework enabled us to evaluate how lag streamflow, cluster specific pretraining, and transfer learning influence hydrological model prediction in both data-scarce and ungauged basins in target region. Groups are defined to evaluate the influence of input predictors such as in Group-I (lag-1, lag-2, lag-3 streamflow), Group-II (temperature, precipitation, PET), and Group-III (combined lagged streamflow, temperature, precipitation, PET). PCA was used for dimensionality reduction with K-Means clustering algorithm to group catchments into five clusters based on static attributes of basins. LSTM models are trained separately for each cluster ($LSTM_{1,2,3}$ & $LSTM_5$) and for all basins combined ($LSTM_6$). Cluster 4 serves as the Base Model ($LSTM_4$) which is regionally trained on target basins of Central Asia (CA). In Experiment 1, pretrained cluster specific LSTMs are fine tuned on all 36 basins in the target region to compare TL-LSTMs with based model ($LSTM_4$). In Experiment 2, the target basins split into two spatial groups (18 for fine-tuning and 18 withheld as ungauged basins). TL-$LSTM_s$ finetuned on 18 basins are used to predict streamflow in the remaining 18 basins (e.g., treated as ungauged) in the target region.

cover (Du et al., 2023; Jehn et al., 2020). To explore the structure of catchment descriptors and identify hydrologically meaningful groups of basins, we applied Principal Component Analysis (PCA) to reduce the dimensionality of the basin descriptor data set. PCA was chosen because it effectively transforms a large set of potentially correlated hydro-environmental variables into a smaller set of uncorrelated principal components while retaining majority of variance in the data set. Catchment descriptors were standardized using z-score normalization to ensure all variables contributed equally to the clustering process, eliminating the influence of differing units and scales. This step transformed the feature set to have a mean of zero and a standard deviation of one. The number of principal components retained for subsequent clustering was determined based on the cumulative explained variance, ensuring that at least 90% of the total variance was captured. This approach reduces computational complexity and minimizes redundancy while preserving essential information.

K-Means clustering algorithm group basins into distinct clusters based on similarity, uncovering patterns and differences among them using the PCA-reduced feature space (Sinaga & Yang, 2020). For basin grouping, we applied the K-means clustering algorithm to the PCA-transformed data. K-means was selected for its simplicity and effectiveness in partitioning data into distinct clusters based on similarity. By performing clustering in the reduced PCA space, we ensure that the algorithm focuses on the most informative patterns in the data while avoiding issues associated with high-dimensional data sets. Silhouette Score provides valuable guidance in selecting the optimal number of clusters. To select optimal number of clusters, the Silhouette Score was computed (Figure S1 in Supporting Information S1). This metric measures the similarity of each point to its assigned cluster

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Table 1
Overview of Data Sets, Variables, and Model Configurations, Including the Training, Validation, and Testing Approaches for Basins, Input Data Groups, Temporal and Spatial Basin Divisions, Donor and Target Basins, and Different LSTM Model Variants

Groups	Variables	Basins	Category	Training strategy	Validation and test strategy	LSTM models
Preliminary Experiments						
Group-I	1, 2- and 3-day lag observed streamflow	GB-SCT, CH, BC and CA	Global catchments	Temporal splitting	Temporal splitting	4-LSTM models
Group-II	pre, temp and pet	GB-SCT, CH, BC and CA	Global catchments	Temporal splitting	Temporal splitting	4-LSTM models
Group-III	1, 2- and 3-day lag observed streamflow, Pre, Temp, PET	GB-SCT, CH, BC and CA	Global catchments	Temporal splitting	Temporal splitting	4-LSTM models
Transfer Learning Experiments based on clusters						
Cluster 1	1-day lag observed streamflow, pre, temp and pet	CH-a	Donor catchments	Donor + Target catchments	Target catchments	$LSTM_1$
Cluster 2	1-day lag observed streamflow, pre, temp and pet	GB-SCT	Donor catchments	Donor + Target catchments	Target catchments	$LSTM_2$
Cluster 3	1-day lag observed streamflow, pre, temp and pet	CH-b	Donor catchments	Donor + Target catchments	Target catchments	$LSTM_3$
Cluster 4	1-day lag observed streamflow, pre, temp and pet	CA	Target catchments	Target catchments	Target catchments	LSTM4
Cluster 5	1-day lag observed streamflow, pre, temp and pet	BC	Donor catchments	Donor + Target catchments	Target catchments	LSTM5
Cluster 6	1-day lag observed streamflow, pre, temp and pet	CH-a, GB-SCT, CH-b and BC	Donor catchments	Donor + Target catchments	Target catchments	$LSTM_6$
Prediction at Ungauged basins						
Cluster 4	1-day lag observed streamflow, pre, temp and pet	CA	Target catchments	Donor + 50% Target catchments	50% Target catchments	$LSTM_{1-6}$

relative to other clusters. For visualization purposes, the first three principal components were used to generate 3D plot of clusters, providing a clear representation of basin groupings and underlying hydro-environmental differences (Figure 3).

Figure 3 illustrates a distinct separation of clusters in the three-dimensional PCA space, indicating that the selected features effectively differentiate the groups. The plot indicates that the clustering algorithm successfully captures the underlying differences in basin characteristics, effectively grouping basins with similar traits. For example, Cluster 2 (blue) is located at higher values of PCA 3, distinctly separated from other clusters. CH region being split into two distinct groups such as Clusters 1 (purple) and Cluster 3 (green) which appear well separated along PCA 1 and PCA 2 axes. The first group of CH basins (e.g., CH-a) represents temperate, continental lowland basins dominated by urban and cropland land use with snowmelt and rainfall-driven runoff, while the second group includes high mountain alpine basins influenced by both snow and glacier melt.

We excluded hydrological response variables from the exploratory analysis to assess hydrological similarity when clustering basins. The resulting clusters were later incorporated as features in predictive models to avoid data leakage. This strategy enhances model robustness and generalizability, particularly in regionalization studies, where the aim is to group basins based on their intrinsic characteristics rather than observed flow responses. Figure 4 presents the basin attributes for each cluster, with all values scaled between 0 and 1 through normalization. Aridity serves as a clear distinguishing factor, separating the humid basins in Clusters 1–3 from the distinctly arid basins in Cluster 4 (Central Asia), which are also distinguished as being the highest in elevation, higher potential evapotranspiration (pet), higher pasture extent (pst) while lowest aridity, lowest soil water content (swc), lowest climate moisture index (cmi) and urban area (urb). Cluster 1 (CH-a) is mainly based on lowland urban (urb) and cropland area (crp), higher population density (ppd), higher actual evapotranspiration (aet). Cluster 3 (CH-b) captures "montane Switzerland" having highest glacier and permafrost area, lowest cropland area, lowest urban area, steepest slope (slp), highest precipitation (pre), highest snow cover area (snw), highest

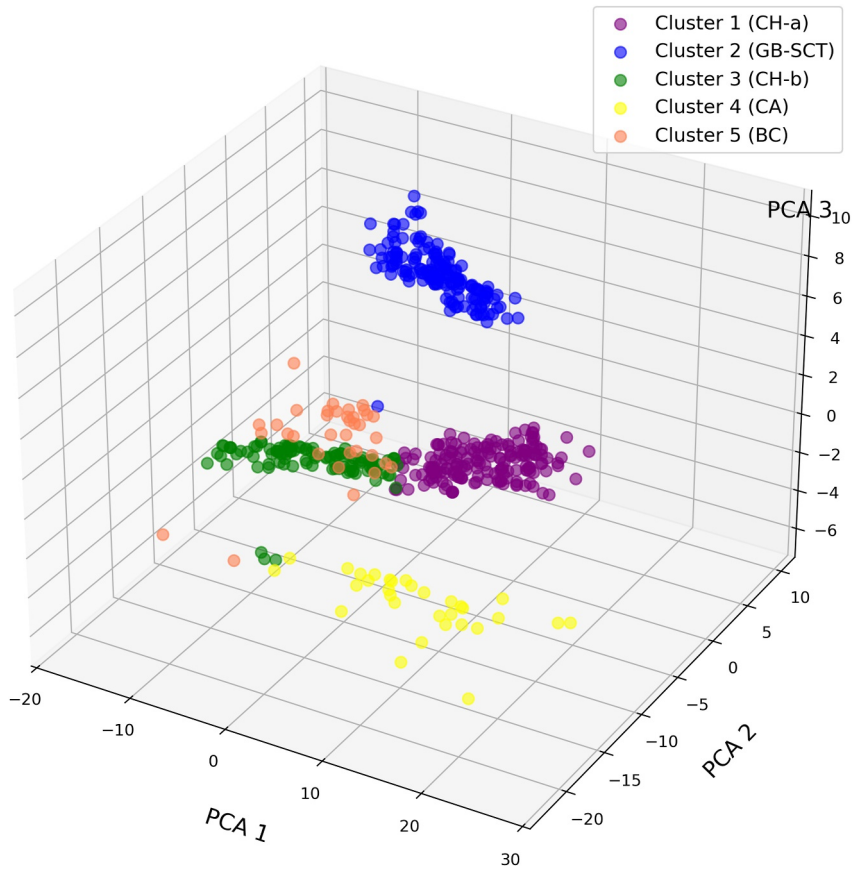


Figure 3. Clustering of basins visualization with 3D plot with first three principal components (PCA 1, PCA 2, and PCA 3) after dimensionality reduction. Each data point represents a basin color coded to reflect its donor region. Cluster 1 (purple) and Cluster 3 (green) are from Switzerland (CH-a and CH-b, respectively); Cluster 2 (blue) is from Scotland (GB-SCT); Cluster 4 (yellow) is from Central Asia (CA); Cluster 5 (brown/coral) is from British Columbia, Canada (BC).

climate moisture index (cmi) and highest soil water content (swc). It indicates that CH basins were subdivided mainly based on glacier and permafrost basin characteristics.

2.5. Long-Short Term Memory (LSTM) Networks

Long Short-Term Memory (LSTM) algorithms integrate both dynamic and static inputs, capturing time-varying features like climate variables (e.g., pre, temp and pet. alongside intrinsic catchment attributes such as drainage area, slope, lithology, and soil properties. LSTM is an improved variant of established Recurrent Neural Network (RNN) modeling schemas offering accurate forecasting for multivariate time series data due to its long-term memory capabilities (Feng et al., 2021). It differs from standard recurrent networks by incorporating cell states and gates, rather than relying on a single state variable (Hochreiter & Schmidhuber, 1997). The *input, forget,* and *output gates* regulate the flow of information by determining which data to retain, discard, or pass on (Figure 5). By leveraging *memory states* and *gating mechanisms*, the LSTM is specifically engineered to handle sequential data and effectively capture long-term dependencies. We implemented LSTM model using Neural Hydrology package (Kratzert et al., 2019).

Hyperparameters like l*earning rate, dropout rate,* and *hidden size* are adjusted manually for optimal performance of the model. A GPU-accelerated architecture (cudalstm) with adjustable weights and nonlinear activations, is used to predict streamflow. LSTM has a basic architecture consisting of *multiple layers, weights, and nonlinear activation functions*. During the training process, the information from the training data is stored as *weights*, with different parts of the network self-organizing to perform specific tasks based on the architecture (e.g., in a streamflow prediction LSTM, some of the *cell states* may track accumulated snow storage) (Lees et al., 2022). An LSTM layer with *128 units* enables the model to capture long-term dependencies in sequential data. The dynamic

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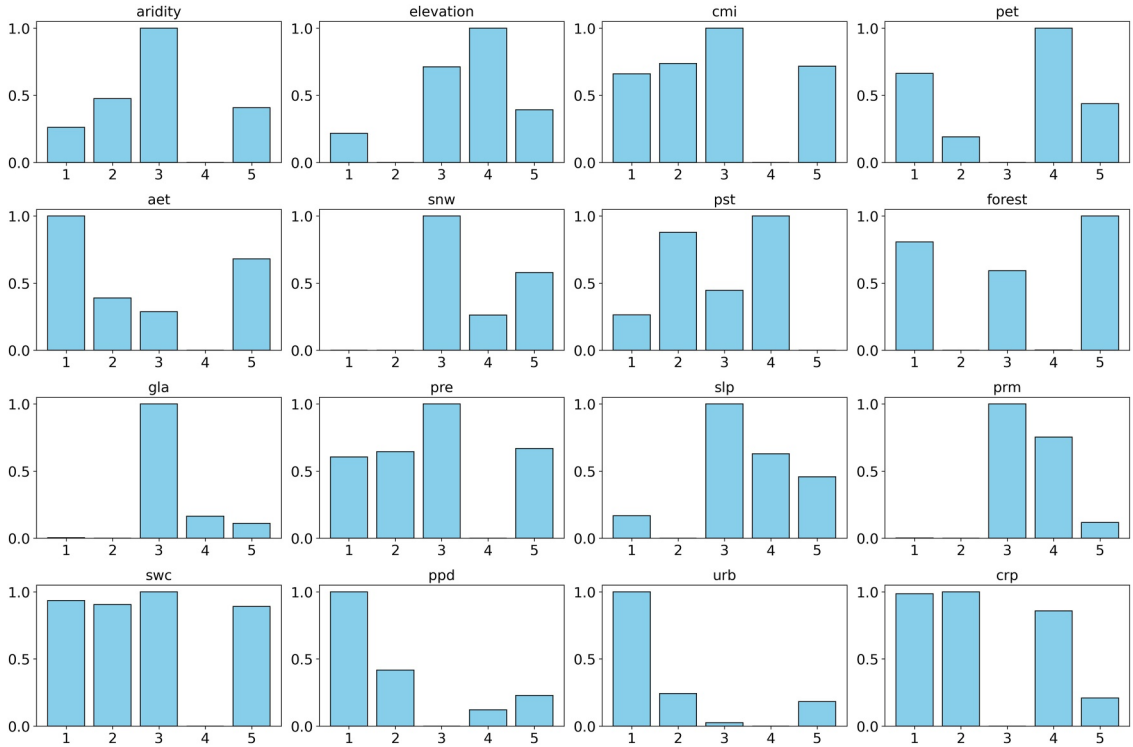


Figure 4. Contribution of selected attributes (normalized data) distinguishing five clusters (1–5) across four regions (CH, GB-SCT, BC and CA). Each subplot represents a variable, with the *x*-axis indicating the number of clusters (1–5) and the *y*-axis showing the attribute values. (i) Aridity index (aridity) (ii) elevation (iii) climate moisture index (cmi) (iv) potential evapotranspiration (pet) (v) actual evapotranspiration (aet) (vi) snow cover (snw) (vii) pasture extent (pst) (viii) forest (ix) glacier area (gla) (x) precipitation (pre) (xi) steepest slopes (slp) (xii) permafrost area (prm) (xiii) soil water content (swc) (xiv) population density (ppd) (xv) urban area (urb) (xvi) cropland area (crp).

inputs (e.g., temp, pre, pet and lag streamflow) feed into single LSTM layer followed by a *dropout layer* with a rate of 0.3 to mitigate overfitting. The *forget gate* is initialized with a bias of 3 to facilitate retention of initial information during early training. The *prediction head* is a regression layer with a *linear activation function*, suitable for continuous streamflow predictions. For one day ahead streamflow simulation, a 365-day *look-back window* is used. This block of 365 days and four variables is subsequently fed into single LSTM layer containing *256 units*. The architecture is optimized using the Adam optimizer and NSE as the *loss function* to minimize hydrological prediction errors. This robust configuration balances complexity and computational efficiency, tailored for large-scale hydrological data sets. This approach builds on the Neural Hydrology library by Kratzert et al. (2019) who experimented with various LSTM model architectures. Their findings show that highest

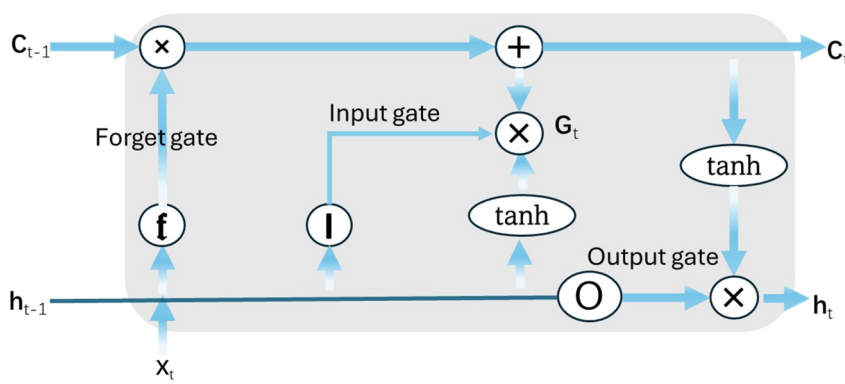


Figure 5. The LSTM unit structure showing the flow of information from forget gate, input gate and output gate. C_t, h_t and X_t are the memory state, hidden state and data fed to cell at time t.

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median performance was shown by a *single-layer* LSTM model of *256 units, dropout rate of 0.4 and an input sequence of 365 days*.

Data were pre-processed and normalized to a range of 0 to 1 ensuring balanced functioning. LSTM models were trained based on input variables, individual basins and clusters (1–5). A series of experiments were conducted to explore the applicability of LSTM models trained in data-rich regions (20 years training period) to data-scarce basins in CA, where only limited data for training LSTM model were available (i.e., training period varies from 1 to 5 years in the target region).

2.6. Model Setup, Training, Test and Validation

To assess the impact of lagged streamflow observations, we employed a temporal splitting strategy, training LSTM models using separate training, validation, and testing periods across the 1980–2015 timeframe for each region individually. For CH, GB-SCT, and BC, the training period spanned 1980–2000, validation covered 2000–2007, and testing used data from 2008 to 2015. In contrast, due to limited streamflow observations in CA, the training, validation, and testing periods varied significantly, ranging from 1950 to 1990. Each CA basin had varying training years (e.g., ranging from 2 to 5 years), while 2 years were consistently allocated for validation and 2 years for testing, depending on data availability. Subsequently, TL experiments adopted a spatial (basin-based) approach, training models on data from donor basins (regions with abundant data) and testing/validating them on target basins (CA) (Table 1).

To incorporate antecedent hydrological conditions, we used lagged streamflow as an additional input feature following established practices in LSTM-based hydrological modelling. We tested three configurations using 1-day, 2-day, and 3-day lagged streamflow (i.e., Q(t–1), Q(t–2), Q(t–3)). Although the LSTM processes input sequences using a 365-day sliding window, this temporal context applies only to the meteorological forcings. The discharge variable was included exclusively through the selected lagged streamflow columns, and no additional historical streamflow values beyond the 1–3-day lags were introduced or zero-filled.

Fine-tuning is the process of taking a pre-trained model and continuing its training on a different but related data set to allow the model to adapt to local characteristics while retaining knowledge from earlier training (Pan & Yang, 2010). For prediction at ungauged basins, target region was divided into two independent groups of 18 basins each. One group was used for fine-tuning, while the other was reserved exclusively for validation. This design ensures that fine-tuning relies only on limited data from a subset of basins, reflecting a data-scarce scenario, while the second group of basins, which the model never sees during training or fine-tuning, represents truly ungauged conditions. Thus, although fine-tuning uses historical data from some target basins, the predictive performance is evaluated on a separate set of basins with no local data, directly addressing the applicability of transfer learning in ungauged or data-limited regions.

2.7. Evaluation Metrics

A comparative analysis was performed using evaluation metrics from hydrological models: Nash-Sutcliffe efficiency (NSE) and Kling-Gupta (KGE) (Gupta et al., 2009; Nash & Sutcliffe, 1970).

$$\mathrm{NSE} = 1 - \sum_{t=1}^{T} \left(Q_{\mathrm{sim}}^{t} - Q_{\mathrm{obs}}^{t}\right)^2 / \sum_{t=1}^{T} \left(Q_{\mathrm{sim}}^{t} - \overline{Q}\right)^2 \tag{1}$$

where Q_{obs}^{t} and Q_{sim}^{t} correspond to the observed and simulated streamflow, respectively, at time t; $\overline{Q}$ is the observed streamflow. The Nash-Sutcliffe Efficiency (NSE) is a commonly used metric to evaluate the predictive accuracy of hydrological models, ranging from $-\infty$ to 1, with a value of 1 indicating perfect correlation.

The Kling-Gupta Efficiency (KGE) is a metric used to assess the performance of hydrological models. KGE values closer to 1 indicate a better model performance, with values less than 1 indicating poorer performance. The equation for KGE is:

$$\mathrm{KGE} = \sqrt{(r-1)^2 + (\beta-1)^2 + (\gamma-1)^2} \tag{2}$$

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where:

r is the correlation coefficient between predicted and observed values.

β is the ratio of standard deviations of the predicted and observed values.

γ is the ratio of the means of the predicted and observed values.

3. Results

3.1. Impact of Lagged Streamflow on Model Performance

Lagged streamflow can add significant value to data-driven deep learning streamflow generation models (Mangukiya et al., 2023). Figure 6 illustrates the performance of basins in each region as evident from boxplots and cumulative density function (CDF). LSTM model showed poor performance in GB-SCT (NSE < 0.5) when trained using 1-day lagged streamflow observations compared to CH (median NSE = 0.703), BC (median NSE = 0.921) and CA (median NSE = 0.76) regions. LSTM model also struggles by using only meteorological data to simulate streamflow in BC and CA regions (e.g., as shown by Group-II). However, when LSTM model trained with 1-day lagged streamflow observations as an additional variable to meteorological data, it increased the prediction performance (median NSE = 0.703 to 0.837) in CH region but no significant improvement in GB-SCT region as evident in median NSE (NSE = 0.84 to 0.85). These results indicate that meteorological variables (e.g., Group-II) are far more effective predictors compared to lagged streamflow alone in the GB-SCT region. However, 1-day lagged streamflow observations alone were sufficient for accurate streamflow predictions in BC region and the inclusion of meteorological variables did not lead to any significant improvement (median NSE = 0.921 to 0.923) here. Specifically, CDF plot illustrates that the LSTM model trained with Group-II and Group-III variables consistently achieved better performance (such as higher NSE values) than the LSTM model trained with only Group-I data.

In terms of the percentage of basins in each region, the LSTM model trained with Group-I data performed well in CH basins, with NSE values remain above 0.5 in 76.5% of the basins. In contrast, GB-SCT region showed a lower performance, with only 48.2% of the basins achieving NSE values greater than 0.5. In BC and CA regions, the model trained with Group-I data demonstrated significant improvements, with 90.3% of basins in BC and 77.8% in CA achieving NSE values above 0.5. When meteorological variables alone (Group-II) were used for training LSTM, performance was moderate, with 67.7% of basins in BC and 66.7% in CA showing NSE values greater than 0.5. However, combining 1-day lag streamflow data with meteorological variables (Group-III) enhanced the performance of LSTM, yielding NSE values above 0.5 in 93.6% basins in BC and 77.8% basins in CA. These results indicate that using the Group-III variables led to improved streamflow predictions compared to training the LSTM with the Group-I and Group-II data sets.

3.2. Comparison of LSTM Training Approaches in Central Asia

After conducting initial experiments to evaluate the impact of lag streamflow on prediction accuracy, Group-III variables were adopted as forcing data for further analysis. LSTM models were trained, tested and validated in target basins using two approaches: locally trained LSTM ($LSTM_{local}$) (e.g., LSTM models trained individually on each basin in CA) and regionally trained LSTM ($LSTM_{regional}$) (e.g., single LSTM model trained on all basins from CA). Both $LSTM_{local}$ and $LSTM_{regional}$, exhibited varied performance across basins, with NSE values ranging from very poor (NSE $\leq$ 0) to excellent (NSE > 0.8) (Figure 7). $LSTM_{regional}$ model shows good generalization compared to $LSTM_{local}$ models by performing better in central, eastern and northern basins of the target region.

The performance metrics of individual basins showed considerable variation between the two training approaches in the target region (CA) such as $LSTM_{regional}$ and $LSTM_{local}$. $LSTM_{regional}$ model training showed poor performance in five basins (Basin ID: 2217400, 2240101, 2217470, 2217410, 2217120, and 2217104), with NSE and KGE values remain consistently below zero, while $LSTM_{local}$ training showed poor performance in eight basins (Basin ID: 2217400, 2240101, 2217470, 2217410, 2217120, 2217104, 2217480, and 2217401). In majority of basins (21 out of 36), the $LSTM_{local}$ training approach showed NSE higher than $\geq$0.5 (Figure 8). These results indicate that $LSTM_{regional}$ outperformed $LSTM_{local}$, achieving a median NSE of 0.77 and KGE of 0.77, whereas the $LSTM_{local}$ approach had a median NSE of 0.65 and KGE of 0.68.

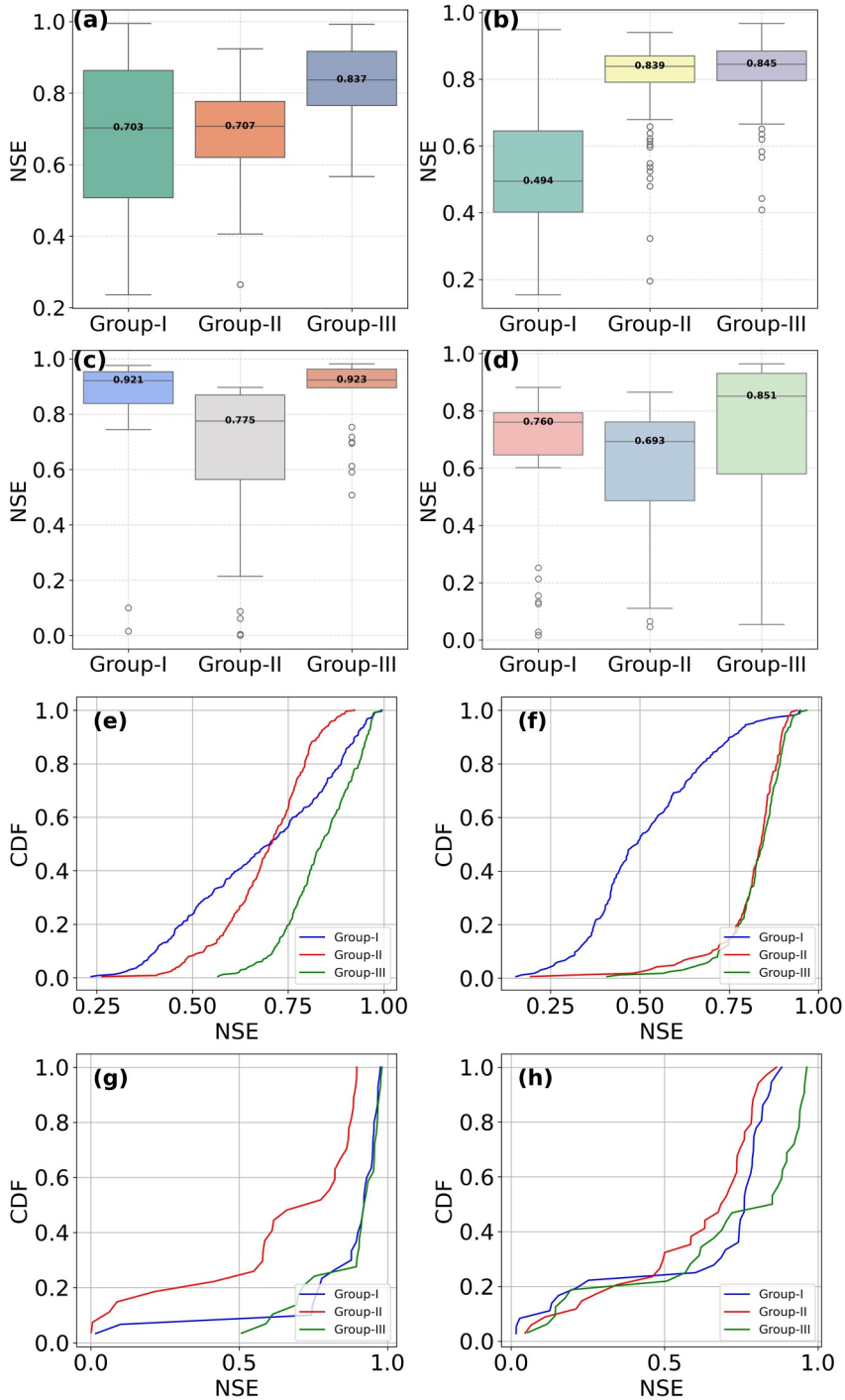


Figure 6. Regional LSTM model results using 1-day lagged streamflow observations and meteorological variables on streamflow prediction performance represented by NSE. Both boxplot (a) Switzerland (CH), (b) Scotland - in Great Britain (GB-SCT), (c) British Columbia in Canada (BC), and (d) Central Asia (CA) and cumulative density function (CDF) (e) Switzerland (CH), (f) Scotland—in Great Britain (GB-SCT), (g) British Columbia in Canada (BC), and (h) Central Asia (CA) illustrate the prediction accuracy of LSTM models using 1-day lagged streamflow observations (Group-I), meteorological variables (Group-II), and by combination of lagged streamflow and meteorological variables (Group-III) to train the model.

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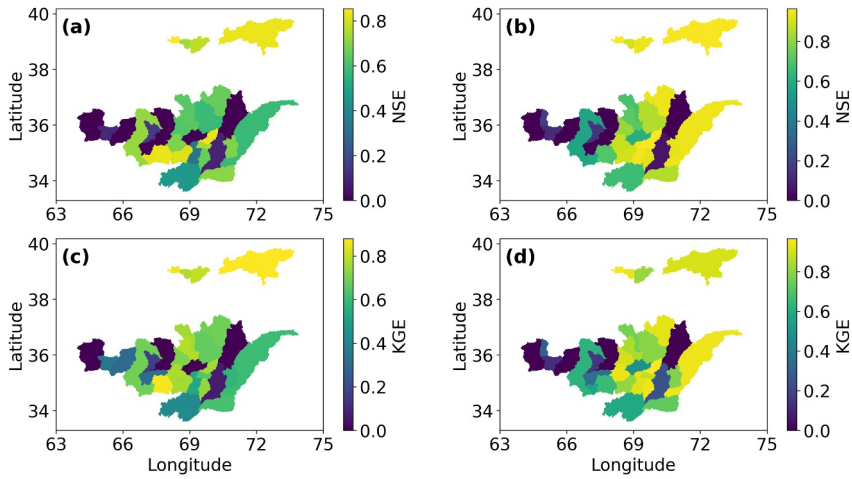


Figure 7. Evaluation of $LSTM_{local}$ and $LSTM_{regional}$ model performance in Central Asia (CA) with NSE and KGE metrics. Subplots show NSE for (a) $LSTM_{local}$, (b) $LSTM_{regional}$ and KGE for (c) $LSTM_{local}$, and (d) $LSTM_{regional}$.

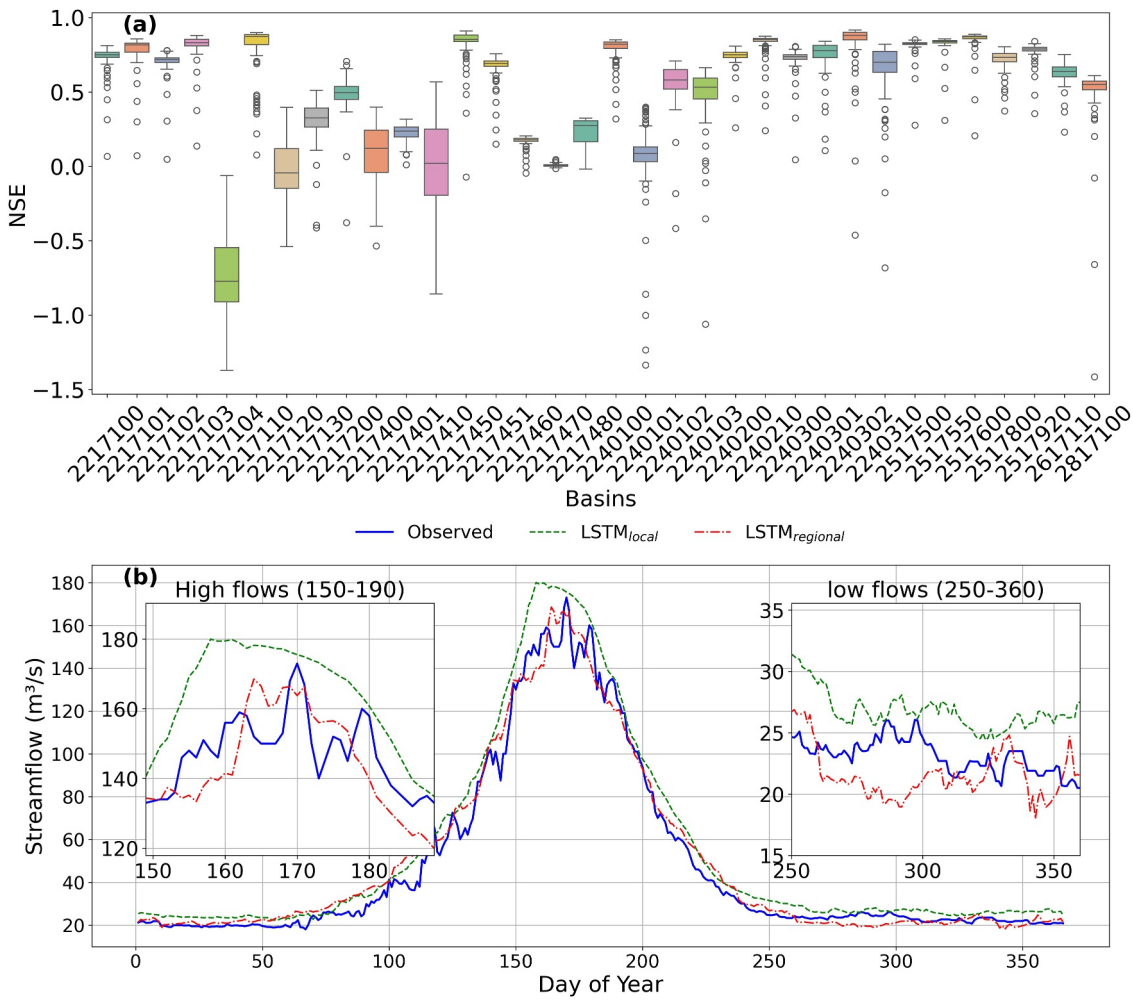


Figure 8. Performance comparison of training approaches in the target region such as $LSTM_{local}$ and $LSTM_{regional}$. (a) The $LSTM_{local}$ model, trained individually on each basin (b) Mean annual hydrographs of $LSTM_{local}$ and $LSTM_{regional}$ and inset maps for high flows (75th percentile) and low flows (25th percentile).

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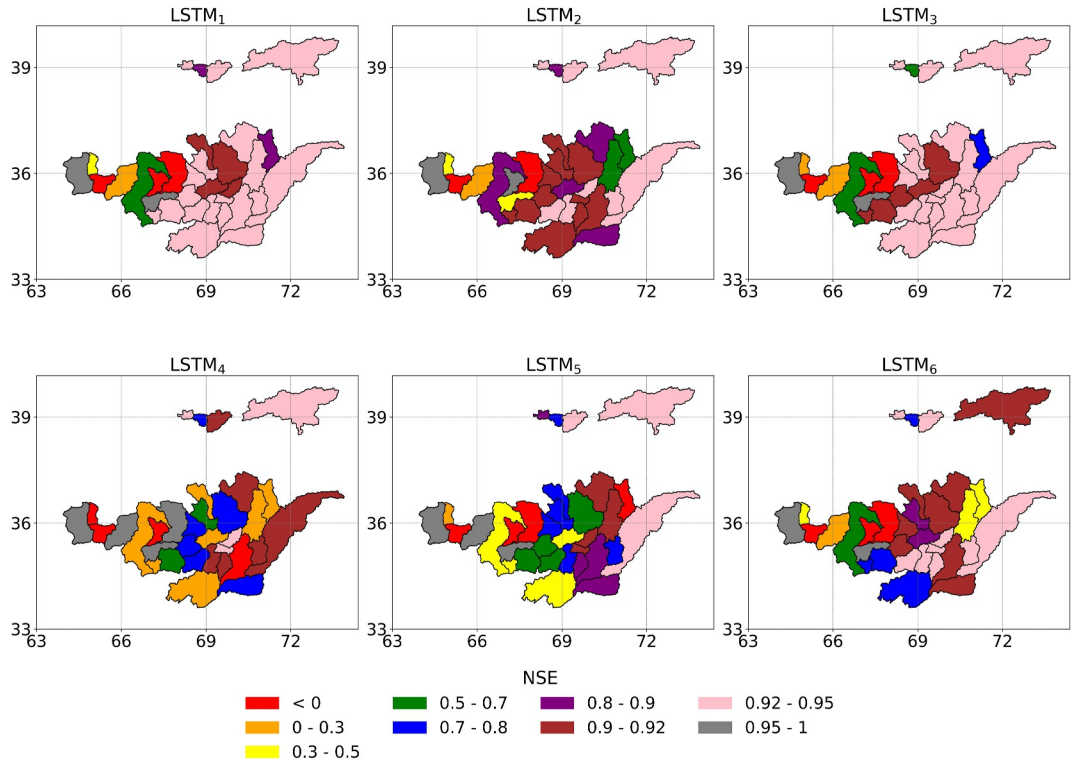


Figure 9. Spatial distribution of basins comparing the performance of transfer learning LSTM models ($LSTM_1$, $LSTM_2$, $LSTM_3$, $LSTM_5$, $LSTM_6$) against the $LSTM_{regional}$ model ($LSTM_4$) across the target basins. The map highlights basins where transfer learning models achieve higher streamflow prediction accuracy than $LSTM_{regional}$ based on the NSE metric.

The prediction accuracy of $LSTM_{local}$ and $LSTM_{regional}$ models was evaluated at the 25th percentile (representing low flows) and the 75th percentile (representing high flows) to compare their performance in simulating low and high flow conditions, respectively. The results are shown by mean daily hydrographs averaged over testing period for both the $LSTM_{local}$ and $LSTM_{regional}$ which show clear differences in their performance during high and low flows (Figure 8). At 25th percentile, both $LSTM_{regional}$ and $LSTM_{local}$ models failed to predict low flows accurately as indicated by underestimation and overestimation of low flows respectively. For high flows, $LSTM_{regional}$ (NSE = 0.94, KGE = 0.94) provided more accurate streamflow predictions compared to $LSTM_{local}$ (NSE = 0.77, KGE = 0.83). $LSTM_{regional}$ model effectively captures the overall trends in high flows, closely matching the observed streamflow. In contrast, $LSTM_{local}$ model tends to overestimate both high flows. Although the $LSTM_{regional}$ model slightly underestimates high and low flows, but it provides a better overall match to the general flow patterns compared to the $LSTM_{local}$ model.

3.3. Performance of Transfer Learning Models

Figure 9 illustrates the performance comparison of the TL-LSTM models (LSTM1, LSTM2, LSTM3, LSTM5, and LSTM6) alongside regionally trained LSTM model ($LSTM_4$) across the target basins. For instance, $LSTM_{regional}$ ($LSTM_4$) performed poorly in seven out of the 36 CA basins, yielding negative NSE values (NSE < 0) for Basin IDs: 2217400, 2217401, 2217104, 2217460, 2217470, 2217480, and 2217410. However, its performance was improved dramatically in two individual basins using TL approach. For instance, NSE and KGE values significantly improved (e.g., NSE = 0.18 to 0.65and KGE = 0.31 to 0.58) in Basin ID: 2217400. Similarly, an increase in NSE (−0.18 to 0.41) and KGE (−0.003 to 0.46) was observed in Basin ID: 2217480 when trained with cluster 1 (CH-a) basins ($LSTM_1$). However, in the remaining five basins (Basin ID: 2217401, 2217104, 2217460, 2217470 and 2217410), there was no significant improvement even with TL-LSTM models.

Most of the basins in the target region show higher or better performance using $LSTM_1$ and $LSTM_3$ (NSE > 0.92) compared to $LSTM_{regional}$ (NSE < 0.89). Both $LSTM_1$ and $LSTM_3$ models demonstrate improved performance in the northern, eastern, and central parts of CA, whereas the $LSTM_4$ model exhibits a more uniform performance

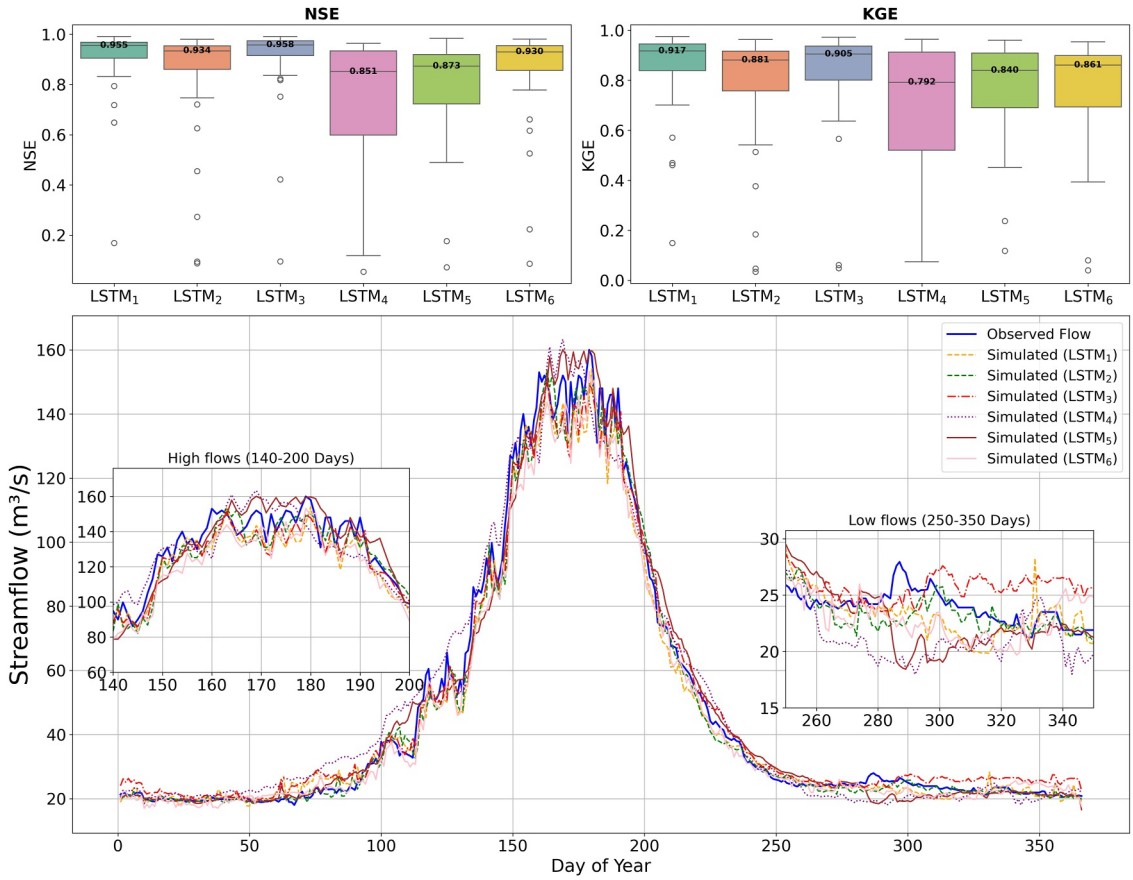


Figure 10. Performance comparison of transfer learning models ($LSTM_1$, $LSTM_2$, $LSTM_3$, $LSTM_5$ and $LSTM_6$) and the $LSTM_{regional}$ model ($LSTM_4$) in the CA. The boxplot presents the variability in streamflow prediction accuracy, highlighting median performance metrics for each LSTM model. The hydrograph visualizes peak and baseflow simulations, comparing the performance of transfer learning models ($LSTM_1$, $LSTM_2$, $LSTM_3$, $LSTM_5$, and $LSTM_6$) against the $LSTM_{regional}$ model ($LSTM_4$).

distribution across the target basins. These results indicate that the TL-LSTM models effectively transfer general hydrological knowledge from the donor catchments to the target basins, enhancing prediction performance in the target region.

Among the TL-LSTM models, the $LSTM_3$ achieved the highest performance, as indicated by higher median NSE = 0.96. However, $LSTM_6$ model pre-trained with a complete set of 441 basins led to a slight decline in performance compared to $LSTM_1$ and $LSTM_3$ models. $LSTM_4$ model showed higher variability and inconsistency with lower performance median metrics NSE = 0.85 compared to TL-LSTM models. These results indicate that relying solely on a smaller, localized data set results in less stable performance in contrast to TL- models which were pretrained on large data sets.

Figure 10 shows the comparison TL-LSTM models with $LSTM_4$ using daily observed and simulated streamflow averaged over the testing years. Streamflow simulations with $LSTM_1$ model closely aligned with the observed high and low flows. Both $LSTM_1$ and $LSTM_3$ demonstrate more accurate representation of high flows. However, $LSTM_2$ and $LSTM_4$ show significant deviations, particularly during high flows. $LSTM_1$ model achieved enhanced simulation accuracy but slightly underestimation of high and low flows. Interestingly, the performance of $LSTM_1$ and $LSTM_3$ trained exclusively on basins from Cluster 1 (160 basins) and Cluster 3 (88 basins) performed better than $LSTM_6$ which was trained using 441 basins (Cluster 1 to 3 and 5). These results indicate that LSTM models trained on sufficiently large and relevant basins (Cluster 1 and 3) exhibit better generalizability and produce more robust streamflow predictions when fine-tuned on target basins.

To examine the impact of training data volume contributions on TL, different strategies were applied in which LSTM model was trained on either a subset of donor basins or the complete data set from each cluster, and

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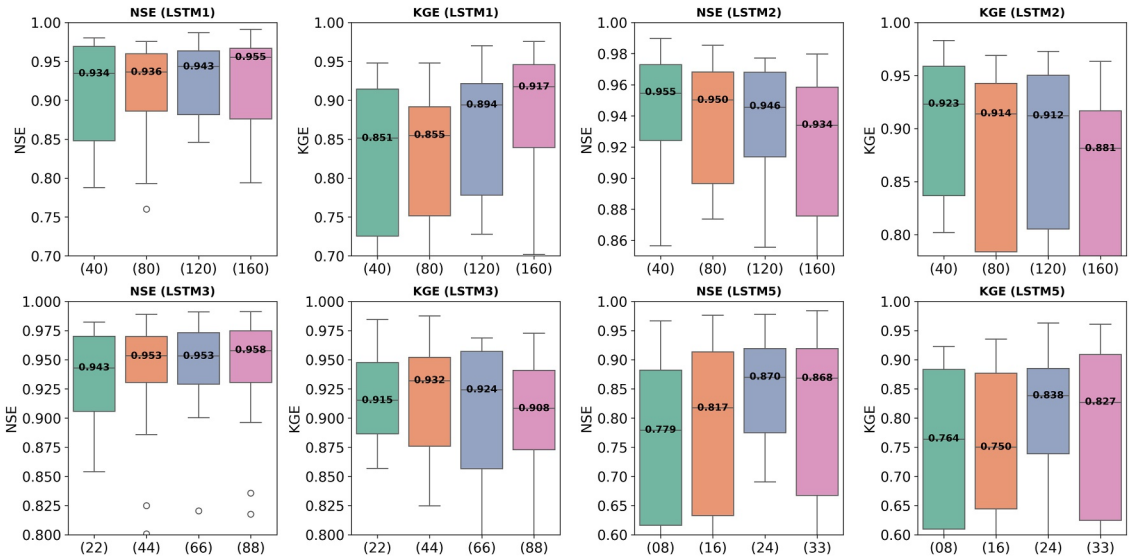


Figure 11. Comparison of TL-LSTM models ($LSTM_1$, $LSTM_2$, $LSTM_3$ and $LSTM_5$) trained with varying numbers of basins from donor region indicated by performance metrics (NSE and KGE). The boxplot illustrates the variability in streamflow prediction accuracy with median performance metrics for each model trained with a quarter, a half, three-quarters, and the total number of basins in each donor Cluster (1–5).

subsequently transferred to target basins for prediction. The impact of the number of basins used for training LSTMs is evident, as illustrated in Figure 11. $LSTM_1$, trained with 160 basins, and $LSTM_3$, trained with 88 basins, demonstrated comparable performance. In $LSTM_1$, increasing the training basins from 40 to 160 improved the median NSE from 0.93 to 0.96 and the median KGE from 0.85 to 0.92. $LSTM_3$ also showed a similar trend, with improvement in performance as the number of basins increased from 22 to 44. However, further increasing the basins from 44 to 66 had negligible effect, with median NSE remaining unchanged while a slight decline in median KGE (0.93–0.92). A further increase in number of basins (e.g., from 66 to 88) resulted in negligible improvement in NSE (0.95–0.96) but a decline in KGE (0.92–0.91). $LSTM_5$ also exhibited performance gains with increase in number of basins from 8 to 16 (NSE increased from 0.78 to 0.82; though KGE declined from 0.76 to 0.75). Moreover, further increase in number of basins (e.g., from 16 to the full data set of 33) improved the performance of $LSTM_5$ (e.g., NSE = 0.82 to 0.87 and KGE = 0.75 to 0.83).

$LSTM_2$ exhibited a decline in performance with increasing training basins. As the number of basins increased from 120 to 160, NSE dropped from 0.95 to 0.93 and KGE from 0.91 to 0.88. A similar decline was observed when increasing basins from 40 to 160, with NSE decreasing from 0.96 to 0.93 and KGE from 0.92 to 0.88. These results suggest that adding more basins may introduce noise or variability, leading to inconsistencies in model predictions, particularly when the new data does not align well with learned patterns. This suggests that the similarity of basins used in TL-LSTM models plays a more significant role in predicting streamflow in the target region than simply the number of basins used for training the LSTM.

Figure 12 provides a comprehensive comparison of the TL-LSTM models and $LSTM_4$ using Cumulative Distribution Functions (CDFs) based on NSE and KGE metrics. In terms of NSE and KGE, the rightward shift of a TL-LSTM models indicates that a larger proportion of basins achieved higher metric values, suggesting better overall model performance against $LSTM_4$. Among the models, $LSTM_3$ consistently stands out as the best-performing TL model. It achieved higher NSE and KGE values across a larger fraction of basins. In contrast, $LSTM_4$ shows a noticeably poorer performance. Its curve is positioned to the left and rises more rapidly, indicating that many basins have lower NSE and KGE values. $LSTM_2$ and $LSTM_5$ perform moderately well, with their curves situated between $LSTM_3$ and $LSTM_4$. They show relatively good performance but not as consistently high as $LSTM_3$. $LSTM_1$ holds a middle-ground position, performing reasonably well but lacking the robustness demonstrated by $LSTM_3$. Its performance is slightly better compared to $LSTM_2$ and $LSTM_5$. $LSTM_6$ shows the poor performance among TL-LSTM models with the lowest NSE and KGE values for most basins, making it the least effective model in this comparison. Overall, the CDF analysis clearly highlights $LSTM_3$ as the most reliable and accurate model for streamflow prediction in the target region.

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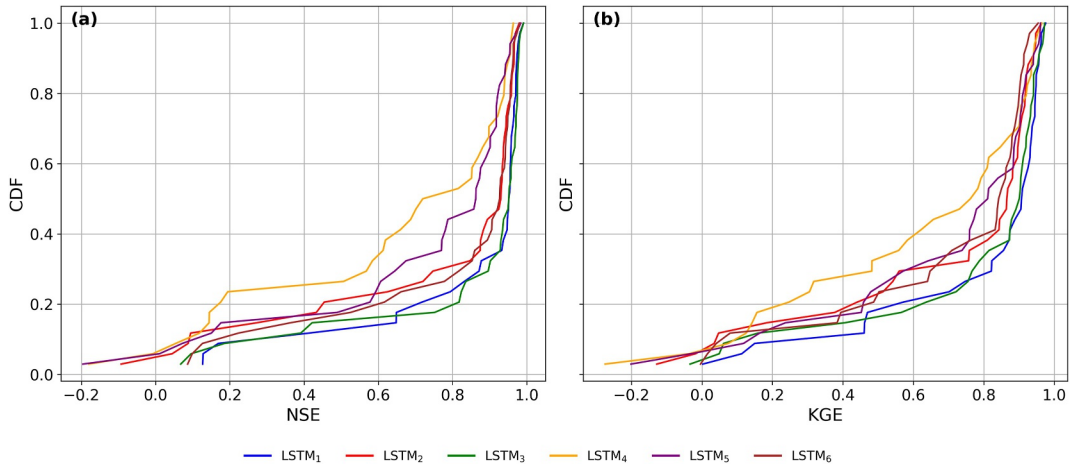


Figure 12. Cumulative Distribution Functions (CDFs) of (a) NSE and (b) KGE values for six LSTM models across multiple catchments.

3.4. Prediction at Ungauged Basins

In the target region (36 basins), streamflow data are limited which restricts their direct inclusion in model training. To examine the performance of TL-LSTM models in ungauged basins, we split the target basins into two groups such as 18 basins for fine tuning, while the remaining 18 (e.g., treated as ungauged) for validation and testing. We evaluated generalization on 18 basins that were never used in training or fine-tuning. The limited available observations in the target basins are used for fine-tuning to adapt the model to local conditions while leveraging knowledge from similar basins. This setup allows the models to leverage knowledge from well-observed basins to predict streamflow under data scarce scenario, without relying on extensive historical records in the target basins.

The performance of the TL-LSTM models ($LSTM_1$, $LSTM_2$, $LSTM_3$, $LSTM_5$, and $LSTM_6$) were evaluated against the $LSTM_4$ model (trained on 18 basins) for streamflow prediction in ungauged basins (Table S4 in Supporting Information S1). $LSTM_1$ demonstrated a balanced performance across the ungauged basins. Though, it performed poorly (NSE and KGE < 0) in five ungauged basins (Basin ID: 2517800, 2217460, 2217410, 2217401, 2217104) while achieved satisfactory ($0 \leq$ NSE and KGE < 0.5) results in another five ungauged basins (Basin ID: 2817100, 2617110, 2217480, 2217470, 2217400). In the remaining basins, $LSTM_1$ model showed excellent performance (NSE and KGE ≥ 0.9) in five ungauged basins (Basin ID: 2240103, 2217210, 2217201, 2217200, 2217120) and moderate to good performance ($0.5 \leq$ NSE and KGE < 0.9) in another three ungauged basins (Basin ID: 2517550, 2217450, 2217451). These results show that $LSTM_1$ model has strong predictive ability for some specific basins. The daily streamflow averaged over the test period indicates that $LSTM_4$ model overestimated high flows in ungauged basins, whereas all other TL-LSTM models demonstrated improved accuracy in predicting both high and low flows in ungauged basins (Figure S2–S7 in Supporting Information S1).

Similarly, $LSTM_3$ model also achieved excellent performance (NSE and KGE ≥ 0.9) in five ungauged basins and moderate to good performance ($0.5 \leq$ NSE and KGE < 0.9) in other three ungauged basins. However, the $LSTM_3$ model struggled in eight ungauged basins, where its performance was very poor (NSE and KGE < 0), and in other two ungauged basins, it achieved satisfactory ($0 \leq$ NSE and KGE < 0.5) results. $LSTM_3$, despite exhibiting higher variability, still achieved strong performance in certain ungauged basins, similar to $LSTM_1$. This suggests that while $LSTM_3$ may struggle with consistency across all ungauged basins, it effectively captures hydrological patterns in some specific ungauged basins.

$LSTM_2$ showed slightly weaker performance than $LSTM_1$. It achieved excellent performance (NSE and KGE ≥ 0.9) in four ungauged basins and moderate results in other two ungauged basins. However, its performance was very poor in seven ungauged basins, where both NSE and KGE remained below zero. $LSTM_2$ displayed higher variability compared to $LSTM_1$ and $LSTM_3$ but was also able to achieve good results in some ungauged basins.

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$LSTM_4$, $LSTM_5$, and $LSTM_6$ consistently showed poor performance across the majority of ungauged basins. $LSTM_4$ performed poorly in 13 out of 18 ungauged basins while achieving excellent performance in only two ungauged basins. Similarly, $LSTM_5$ exhibited very poor performance in 14 basins, with moderate results in two basins and excellent performance in the remaining two basins. Similarly, $LSTM_6$ showed very poor performance in 12 ungauged basins and moderate performance in remaining five ungauged basins. $LSTM_6$ failed to achieve NSE higher than 0.9 in any of the ungauged basins, highlighting its limited applicability.

Among all LSTM models, $LSTM_1$ and $LSTM_3$ trained with clusters 1 and 3 (e.g., CH-a and CH-b), showed the best performance (NSE and KGE ≥ 0.9), indicating their suitability for streamflow prediction in certain ungauged basins. In contrast, $LSTM_4$, $LSTM_5$, and $LSTM_6$ trained with basins or clusters from CA, BC and combined data from all basins respectively, displayed poor predictive capabilities across most of the basins. $LSTM_2$ trained with cluster 2 (e.g., GB-SCT) showed moderate performance, performing well in specific cases but with higher variability. These results indicate that $LSTM_2$ and $LSTM_3$ are the most effective models for predicting streamflow in ungauged basins.

4. Discussion

To systematically evaluate the impact of training strategy and basin similarity on predictive performance, this study evaluated different approaches such as basin-specific LSTMs trained independently for each basin ($LSTM_{local}$), a regionally trained LSTM on multiple basins ($LSTM_{regional}$), cluster-based TL-LSTMs ($LSTM_1$ to $LSTM_5$ except $LSTM_4$) that exploit hydrologically similar groups of basins, and LSTM trained on pooled data set ($LSTM_6$), which combine all available basins without accounting for similarity to leverage cross-basin knowledge. This comprehensive set of variants allows for a detailed comparison of how model architecture and training strategy influence performance of models in data-scarce and ungauged basins.

With recent progress in deep learning, new solutions are emerging to improve runoff prediction in regions with limited data (Xu et al., 2023; Yao et al., 2023). Although traditional process-based models are grounded in physical principles, they often struggle with reliable parameter estimation and showed limited ability to generalize across diverse hydrological conditions (Guo et al., 2021; Perrini et al., 2024). Conversely, purely data-driven models, relying heavily on ML algorithms, demand vast amounts of high-quality training data to achieve satisfactory prediction accuracy (Lu et al., 2021; Sattari & Moradkhani, 2025). However, most catchments around the world have limited or no observations, making direct calibration impossible (Bao et al., 2024). As a result, predicting streamflow in such regions has become a key challenge in hydrology (Ghaneei et al., 2024; Istalkar & Biswal, 2024). Transfer learning (TL) offers a promising solution to these challenges by utilizing knowledge acquired from data-rich domains to enhance model performance in data-scarce domains (Kambale et al., 2023; Khoshkalam et al., 2023). The success of knowledge transfer is directly related to the degree of similarity between these domains (Gerace et al., 2022). LSTM model trained in data-rich regions can enhance predictions by fine-tuning with limited local data, allowing it to adapt to new catchments with differing climate or landscape characteristics (Nearing et al., 2021). Similarly, it has been demonstrated that the use of lagged discharge in LSTM networks can improve streamflow prediction with so-called data integration (DI) approach (Khoshkalam et al., 2023).

In most existing studies, LSTM models are trained within a specific regional context and their ability to generalize is evaluated in similar or neighboring regions (Choi et al., 2022; He et al., 2024; Khoshkalam et al., 2025; Mangukiya et al., 2023; Wilbrand et al., 2023) with only a limited number of studies exist to explore cross-regional model transfer (Ma et al., 2021; Zhang et al., 2024). Following the analogy that catchment attributes help to identify similar basins, models can be more effectively "transferred" from data-rich regions to data-scarce ones (Tang et al., 2025). In this study, we extend that scope by training models on diverse data-rich catchments and evaluating their performance in data limited conditions. Our findings indicate that cross-regional transfer was particularly effective between CH and CA basins, owing to their closely aligned hydrological characteristics. This represents a significant step toward truly global model transferability, moving beyond regional confines. Our results show that TL-LSTMs significantly enhances model generalizability in data-scarce regions with similar hydrological characteristics which highlight the importance of assembling a representative training data set. We also found that both the quantity and heterogeneity of basins in the training clusters are key factors in improving LSTM performance in ungauged basins. By validating TL across vastly different regions, our work provides new insights into the potential and limitations of global-scale hydrological modeling. It opens pathways for more

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universally applicable predictive models that reduce reliance on localized data collection, offering practical benefits for water resource management in data-limited regions of the world.

4.1. Data Integration for Streamflow Prediction

Accurate streamflow prediction depends on capturing both historical flow patterns and meteorological influences. LSTM models can leverage these complementary sources of information, but their relative importance may vary across catchments. To evaluate this, we developed LSTM models for basins including CH, GB-SCT, BC, and CA, and demonstrated that incorporating lagged streamflow observations significantly enhances predictive accuracy, underscoring their critical role in model performance as supported by previous research (He et al., 2024; Khoshkalam et al., 2025; Mangukiya et al., 2023). The inclusion of 1-day lagged streamflow data substantially improves streamflow predictions in CH, BC, and CA, but has limited benefits in GB-SCT basins. The higher performance in CH, BC, and CA basins can be attributed to the dominance of seasonal snowmelt (Figure 4), which create a strong temporal dependency between past and future streamflow. In rainfall-dominated GB-SCT basins, rapid hydrological responses limit the predictive value of lagged streamflow, resulting in poor performance (NSE < 0.5). These basins may have influenced by anthropogenic activities or experience frequent and intense storm events. Feng et al. (2021) previously demonstrated that in systems dominated by flash floods with minimal memory effects, such as those in Texas, USA, LSTM model incorporating lag discharge provides no significant advantage. Additionally, potential anthropogenic influences and frequent storm events in GB-SCT might have contributed to the weak predictive power of lagged streamflow observations. These results suggest that the selection of input variables should be tailored to regional hydrological characteristics to maximize model performance.

The variability in model predictions can offer valuable insights. It captures the potential range of streamflow, which is critical for effective planning. Using a multi-model ensemble of streamflow predictions, we generated a robust range of potential streamflow behaviors across catchments and over time. Our results indicate strong model agreement, particularly during high-flow events (Figure 10), where simulations consistently bound the observed discharge within ±10 cumecs, equating to around ±5% error margin. The ensemble captures key hydrological behaviors, including extremes at both ends of the flow spectrum. This convergence shows that model captures the dynamics of snowmelt and rainfall-runoff processes effectively, making it a robust tool for flood forecasting and early warning systems. However, during low flows, the results are more complex. Although the absolute deviation remains small (e.g., ±5 cumecs), the relative error increases significantly, often ranging from ±25% to ±40%. This increased variability in low flows presents a challenge, particularly in regulatory contexts where minimum environmental flow requirements must be met, or abstraction limits for irrigation are enforced.

4.2. Individual and Regional Approach for Training LSTMs

Several studies have already examined local and regional LSTMs and compared their performance with multiple conceptual models (Arsenault et al., 2023; Gauch et al., 2021; Kratzert et al., 2019; Lees et al., 2021). LSTMs trained regionally particularly on limited or noisy data can lead to overfitting problems (Liu et al., 2023) and outputs may vary significantly in response to small changes in input data (Chaudhary et al., 2022). Our study investigated the key differences in the performance of $LSTM_{local}$ (e.g., basin-specific trained LSTM models) and $LSTM_{regional}$ (e.g., single LSTM model trained on all basins). According to Mangukiya et al. (2023), $LSTM_{regional}$ tend to outperform $LSTM_{local}$. $LSTM_{local}$ trained per basin ensures that localized patterns are captured but the results are inconsistent due to variability in data quality or regional complexity across individual basins. $LSTM_{regional}$ approach helps the model to generalize across different catchments, making it useful for regions with limited data availability (Figure 8).

Hydrological processes vary significantly within and across basins likely due to sparse and insufficient training data, differences in climate and weather patterns, topography, land use and vegetation cover, and snow and glacial melt contributions. $LSTM_{local}$ models are often ill-equipped to handle this variability. Some basins may experience extreme hydrological events such as floods and droughts, making it difficult for $LSTM_{local}$ to capture both high and low flows accurately. When catchments share similar characteristics, the chances of achieving successful knowledge transfer are considerably higher (Liu et al., 2024; Wilbrand et al., 2023). As highlighted in previous studies (Mangukiya et al., 2023; Sahu et al., 2021; Yadav & Mangukiya, 2021), these factors create challenges for hydrological modelling and water resource management in such basins. On the other hand,

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$LSTM_{regional}$ can ingest data from all available catchments and learns relationships and patterns from the entire data set. This allows the model to generalize more effectively across diverse catchments, capturing complex hydrological behavior without being constrained by assumptions about similarity.

In our study, most of the poorly performing basins remained same for both $LSTM_{local}$ and $LSTM_{regional}$ models. $LSTM_{regional}$ did not lead to significant performance improvements in these specific basins. Improvements in $LSTM_{regional}$ model was primarily observed in those basins where initial NSE and KGE values by $LSTM_{local}$ was already moderate (e.g., NSE and KGE $\geq$ 0.2). This suggests that the underlying hydrological complexity and data limitations in certain basins hindered model performance, regardless of the training approach used. Further analysis revealed that $LSTM_{regional}$ outperformed $LSTM_{local}$ in both low flow and high flow conditions (Figure 6). These results are consistent with multiple studies (Kratzert et al., 2019; Mangukiya et al., 2023; Zhang et al., 2024) which demonstrated that training models on diverse data sets can lead to improved performance. However, the performance gap between $LSTM_{regional}$ and $LSTM_{local}$ highlights the need for further investigation into improving individual basin-specific models, especially for data-scarce regions with complex hydrological behaviours.

4.3. Impact of Data Quantity and Quality on TL-LSTMs

In traditional regionalization studies, the degree of similarity between donor and target (e.g., target basins) basins plays a key role in determining the accuracy of hydrological models for streamflow prediction (Bao et al., 2024; Oudin, 2004). Our findings highlight the importance of both data set diversity and relevance in enhancing the robustness and generalizability. Both the quantity and diversity of training data set can enhance the performance of LSTMs (Ma et al. (2021), with model accuracy primarily influenced by local hydro-climatic conditions (Zhang et al., 2022). Even with limited data, LSTM model can learn useful relationships that generalize well to a target basin. For instance, $LSTM_3$ model achieved the highest performance in 88 basins, which supplied relevant data enabling the model to generalize effectively to the target region. It can be inferred that basins in Cluster 3 are more closely related to those in the target region (Figure 4). These similarities (or mismatches) are important because a high degree of dissimilarity can reduce model transferability (Hashmi et al., 2022). On the other hand, $LSTM_6$, trained with the largest and most diverse data set (441 basins), demonstrated a slight decline in performance. A large and diverse data set can improve model generalizability, but it can also introduce noise and reduce the predictive performance of model.

By applying K-means clustering to 441 basins, the data set was divided into five distinct Clusters, each representing basins with similar hydrological characteristics (e.g., precipitation, temperature, elevation, land cover). This approach resulted in improved predictive performance, even though it included a relatively smaller number of basins ($LSTM_3$). Similarly, the comparison between $LSTM_3$ and $LSTM_6$ reveals that the quality of the training data is more critical than the quantity (Figure 11). $LSTM_2$ trained on basins dissimilar to target region, forces the model to acquire conflicting hydrological behaviors. Zhang et al. (2024) demonstrated that pretrained model on UK basins performed poorly when transposed to basins in Chile. The poor performance was attributed to the comparative homogeneity of UK basins and asynchronous climate change responses in northern and southern hemispheres (Collins et al., 2024). Similarly, $LSTM_2$ model showed the least improvement in predictions for the target basins because of significant differences between the donor and the target basins. As a result, increasing the number of basins in the $LSTM_2$ model (from 40 to 160 basins) further degraded the ability of model to make accurate predictions in the target region. In big data ML trends, diminishing returns are commonly observed as the marginal benefit of increasing data set size gradually declines at larger sample sizes (Ma et al., 2021; Sun et al., 2011). Such deterioration in performance can likely be attributed to the introduction of noise or variability in the training data, where the additional basins may not align well with the hydrological patterns in the target basins. Therefore, a careful selection of training data in TL-LSTM models is needed, as the number of basins should not be increased indiscriminately.

4.4. Predictions at Ungauged Basins

This study is intended for basins with limited observations, where the model can leverage knowledge from data-rich donor basins and fine-tune predictions using the available local records. The core concept is to leverage gauged basins to pretrain and fine tune models and subsequently transfer this knowledge to ungauged basins. We used 18 basins from the target region for fine-tuning, while the remaining 18 basins (i.e., treated as ungauged)

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were used for prediction based solely on meteorological forcings. This design preserves the core objective of improving predictions in ungauged basins while enabling a controlled assessment of how additional hydrological information enhances transferability.

A major challenge in streamflow prediction at ungauged basins is intricately tied to the variability in forcing data and catchment characteristics. The Kolmogorov–Smirnov (KS) test results indicate statistically significant distributional differences between the training clusters and the 18 ungauged target basins across precipitation, temperature, potential evapotranspiration, and streamflow (Table S3 in Supporting Information S1). Clusters 1, 2, and 6 show consistently high differences across most variables, while Clusters 3 and 4 exhibit moderate-to-high differences. These data distribution mismatches challenge TL, as LSTM models trained on donor basins may struggle to generalize to target basins with unseen conditions. These distributional mismatches between training and testing data sets likely contributed to the variability in LSTM model performance across the ungauged basins. In training data, majority of basins are likely rainfall-dominated while ungauged basins are governed by snow and glacier processes.

Based on the observed hydrographs (Figure S2–S7 in Supporting Information S1), the basins in target region can be broadly grouped into three regime types: (a) snowmelt- and glacier-dominated basins (e.g., Basin ID: 2217200, 2217210, 2517800, 2617110, 2817100), which exhibit low flows in winter, sharp spring rises, and sustained high flows in summer driven by snow and glacier melt; (b) rainfall-dominated basins (e.g., Basin ID: 2217410, 2217401, 2217460, 2217450, 2217470, 2217480), characterized by multiple peaks and flashy responses to precipitation events throughout the year; and (c) transitional basins with both snowmelt and rainfall influence (e.g., Basin ID: 2217104, 2217201, 2240103, 2517550), which display a distinct spring peak but also additional rainfall-driven variability. The failure of the model to capture these diverse hydrological behaviors highlights the limited transferability to target basins. In particular, the inability to represent snow accumulation, melt dynamics, and glacier contributions leads to severe underestimation or misplacement of seasonal peaks. The poor performance observed in certain basins highlights important limitations of TL when applied to hydrologically complex or extreme environments. To mitigate these issues, effective TL strategies should integrate domain adaptation techniques, such as basin-wise normalization, weighted training based on similarity, and fine-tuning with similar target basin observations to better align the statistical properties of donor and target domains. Basins with predominantly low flows (e.g., Basin ID: 2217104, 2217401, 2217410) were poorly simulated, with models either generating spurious flows during dry periods or failing to capture short-lived peaks. Feng et al. (2020) found that while LSTM models performed well in snow-dominated regions, they were less effective in regions with low discharge volumes, such as those dominated by baseflow. Larger basins with heterogeneous physiographic and climatic conditions also showed reduced predictive skill (e.g., Basin ID: 2617110, 2817100), as spatial averaging likely masks localized runoff dynamics. Kratzert et al. (2024) highlighted that LSTM models perform best when trained on data from a large number of basins, as training on small, homogeneous data sets can cause overfitting and poor generalization. Zhang et al. (2024) reported that transferability is constrained when training data lack sufficient hydroclimatic diversity. These findings emphasize that limited data for fine tuning model can lead to less robust and less reliable predictions at ungauged basins (Khoshkalam et al., 2025).

Basin-specific climate and topographic characteristics play a major role TL to predict streamflow in ungauged basins (He et al., 2024; Zhao et al., 2025). In our study, $LSTM_2$ underperformed compared to $LSTM_3$ at ungauged basins, mainly because Cluster 2 (GB-SCT) basins differ hydrologically from the target basins. A clear contrast emerges between basins in Cluster 2 (GB-SCT) and Cluster 4 in terms of hydroclimatic regimes as shown by Taylor plots (Figure 1). For instance, $LSTM_2$ trained with basins exhibits relatively mild temperatures throughout the year, consistently high and evenly distributed precipitation, and comparatively low and more stable pet throughout the year. In contrast, basins in target region are characterized by a continental climate, with pronounced seasonality in temperature (cold winters and hot summers) and substantially higher pet during summer months. These differences highlight the diverse climatic controls shaping streamflow generation in the two regions and underscore the challenges of TL-LSTM models across contrasting hydroclimatic settings. In addition, the topography differs, as basins in Cluster 2 generally exhibit flatter, lower-relief landscapes, whereas those in CA are predominantly mountainous. Previous studies have shown that hydrological similarity of training data has a greater impact on LSTM model performance than the sheer number of basins used for training (Khoshkalam et al., 2023; Ma et al., 2021; Tang et al., 2025).

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4.5. Limitations of Study

Although the model had no direct access to streamflow data from the evaluation basins during training, it benefits indirectly from patterns learned in the fine-tuned basins within the same region. Therefore, we refer to these basins as "ungauged" for the purposes of model assessment, while noting that their predictions are influenced by regional fine-tuning and are not strictly independent in a global prediction at ungauged basins sense.

The weak performance in specific basins underscores the inherent challenges of applying TL to basins with complex or extreme hydrological conditions. To address these limitations, future work could explore hybrid physics-informed ML approaches, where explicit representations of snow accumulation/melt, evapotranspiration, or groundwater storage are integrated with data-driven models. Incorporating remote sensing products (e.g., snow water equivalent, soil moisture, evapotranspiration) as auxiliary inputs may also improve performance in data-scarce and snow-dominated environments. Moreover, designing region-specific or climate-informed pretraining strategies, where models are initialized with data from hydro-climatically similar basins before transfer, may enhance predictive skill compared to relying on homogeneous donor regions. Finally, expanding training data sets to include more diverse basins can help models learn a broader range of hydrological regimes, thereby improving transferability to ungauged basins with complex or extreme behaviors.

The LSTM models in this study rely heavily on lagged streamflow, which may lead to memorization of past flows rather than capturing the underlying rainfall–runoff processes. As a result, the sensitivity of model to meteorological inputs such as precipitation and potential evapotranspiration may be limited, reducing their reliability for assessing future climate-change impacts. Feature importance analysis can be conducted to determine whether the models rely excessively on past discharge, potentially leading to a "memorization" of historical flows rather than capturing the full dynamics of rainfall–runoff processes. Consequently, the sensitivity of the models to changes in climate inputs is reduced, potentially limiting their applicability for assessing climate-change impacts. Future work should explore strategies to balance the influence of lagged discharge and meteorological inputs to enhance model generalization under varying climate conditions.

5. Conclusions

This study evaluate the effectiveness of LSTM-based transfer learning (TL-LSTM) rainfall–runoff modeling in the data-scarce region of Central Asia (CA) by leveraging donor basins from data-rich regions (CH, GB-SCT, and BC). Donor basins were grouped into hydrologically meaningful clusters using catchment attributes to pretrain multiple TL-LSTM models. By comparing base model ($LSTM_4$) with the TL-LSTMs, we evaluated the advantages of TL over a regional model trained solely on the target basins. For predictions at ungauged basins, the TL-LSTM models were fine-tuned using data from 18 basins in the target region, while the remaining 18 basins were reserved for evaluation and testing. Although the methodological components are established techniques, our findings provide generalizable insights that advance understanding of how and under what conditions TL can improve streamflow prediction in data-limited and ungauged conditions.

The impact of lagged streamflow inputs differed across regions. Lagged streamflow significantly improved predictions in glacier- and snowmelt dominated regions (CH, BC, CA) but provided limited performance gains in GB-SCT region. However, combining lagged streamflow with meteorological forcing inputs yielded the best performance which underscore the importance of region-specific input variable selection in hydrological modeling.

The dominant factor in controlling the performance of TL-LSTM models was the catchment similarity between donor and target basins. TL-LSTM models pretrained on clusters having basins that closely resembled target region ($LSTM_3$) consistently outperformed those trained on pooled data ($LSTM_6$) or less similar basins. $LSTM_2$ Although adding more donor basins in training $LSTM_2$ initially improved performance but further increasing the number of basins contributed little or even degraded performance by introducing irrelevant hydrological patterns. These results demonstrate that TL-LSTM can effectively enhance streamflow prediction in data-scarce regions when donor basins are selected strategically based on hydrological similarity. By splitting the target region into two groups (18 basins for fine-tuning and 18 for validation), we tested TL under realistic data-scarce and ungagued conditions. Fine-tuning with limited local data substantially improved prediction accuracy in the "blind-testing" basins treated as ungauged, showing that even short-record data can meaningfully adapt a regional model to local hydrological behavior.

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This study provides generalizable insights for future TL hydrological modeling. Although we used five clusters, the conclusions are not tied to this specific number. The broader insights such as the dominance of hydrological similarity, the value of limited-data for fine-tuning, diminishing returns beyond sufficient donor-basin diversity, and the superiority of cluster-based pretraining are scalable and remain applicable even if future studies derive more (or fewer) clusters from larger training data sets. These principles offer practical guidance for designing TL workflows in other basins worldwide. This work provides a conceptual contribution by identifying the scenarios under which learned hydrological representations can be reliably transferred. These insights advance our understanding of cross-basin generalization and support the development of efficient, scalable modeling strategies to improve hydrological prediction at gauged and ungauged basins under data scarce scenarios.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

We used daily streamflow observations from the Global Runoff Data Centre (GRDC) [http://grdc.bafg.de], which provides long-term discharge records from thousands of river gauges worldwide. Catchment descriptors and meteorological forcing data from ERA5-Land were obtained using the CARAVAN framework [https://github.com/kratzert/Caravan/] (Kratzert et al., 2023). CAMELS-CH data set are freely available at [https://doi.org/10.5281/zenodo.7784632] (Höge et al., 2023) and CAMELS-GB is freely available via the UK Centre for Ecology and Hydrology Environmental Information Data Centre [https://doi.org/10.5285/8344e4f3-d2ea-44f5-8afa-86d2987543a9] (Coxon et al., 2020). We implemented the LSTM model using the NeuralHydrology Python package, which is freely available at https://github.com/neuralhydrology/neuralhydrology (Kratzert et al., 2022).

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